Error Propagation in Downstream NLP Tasks

INNOPOLIS UNIVERSITY

Case Study in Natural Language Processing course, Spring 2026

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Motivation

I chose this project because I was interested in understanding how errors in one NLP component affect the rest of the pipeline. Many language processing systems rely on syntactic parsing, but it is not always clear how sensitive downstream tasks are to parsing mistakes. By experimenting with controlled corruption, I wanted to see how strongly semantic role labeling depends on different types of syntactic structure.

Key features

- Error propagation is gradual, not catastrophic.
- Dependency errors produce higher variance in SRL performance.
- Structural representations affect robustness differently.
- Combining representations improves stability under noise.

Future work

- Use real benchmark datasets.
- Apply neural SRL models.
- Simulate more realistic parsing errors.
- Explore stronger hybrid integration strategies.

Tools

Experimental results

corr	dep mean	dep std	const mean	const std	hybr mean	hyb std
0.0	1.0	0.0	1.0	0.0	1.0	0.0
0.1	0.95	0.12	0.97	0.09	0.96	0.13
0.3	0.81	0.28	0.91	0.16	0.82	0.24
0.5	0.7	0.32	0.83	0.17	0.75	0.29
0.7	0.58	0.36	0.78	0.16	0.62	0.32

Link

https://github.com/Danil1Danil/case_study/tree/main

Conclusions

- SRL performance decreases as syntactic corruption increases.
- Dependency-based SRL is more sensitive to parsing errors.
- Constituency-based SRL shows higher robustness under structural noise.
- The hybrid approach partially mitigates dependency errors but does not outperform constituency parsing in this setup.
- Error propagation significantly affects downstream tasks in NLP pipelines.